

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The Addition Theorem for two mutually exclusive events A and B states: A) $P(A \cup B) = P(A) + P(B)$ B) $P(A \cup B) = P(A) \cdot P(B)$ C) $P(A \cap B) = P(A) + P(B)$ D) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 Conditional probability $P(A|B)$ is defined as (provided $P(B) > 0$): A) $P(A \cap B) / P(B)$ B) $P(A \cap B) / P(A)$ C) $P(A) / P(B)$ D) $P(A) \cdot P(B)$
 A bag has 3 red and 5 blue balls. Two balls are drawn without replacement. $P(\text{both red}) =$ A) $3/28$ B) $3/32$ C) $6/56$ D) $1/8$
 A random variable X is best described as: A) A variable that takes a fixed constant value B) A function that assigns a real number to each outcome in a sample space C) The probability of an event D) A variable with no defined domain
- 5 For constants a and b, $E(aX + b) =$ A) $a \cdot E(X)$ B) $E(X) + b$ C) $a \cdot E(X) + b$ D) $a \cdot b \cdot E(X)$
- 6 The Poisson distribution arises as a limiting case of the Binomial $B(n, p)$ when: A) $n \rightarrow \infty$ and $p \rightarrow 1$ B) $n \rightarrow \infty$, $p \rightarrow 0$ such that $np = \lambda$ remains constant C) n is small and $p = 0.5$ D) $np \rightarrow \infty$
 For a Binomial distribution $B(n, p)$, the mean and variance are respectively: A) np and np B) np and npq where $q = 1 - p$ C) npq and np D) n/p and npq
- 8 Which of the following is a property of the Normal distribution? A) It is positively skewed B) Mean, Median and Mode are all equal and located at the centre C) The curve touches the x-axis at $x = \mu \pm \sigma$ D) The total area under the curve is 2
- 9 Correlation measures: A) Difference between two variables B) Relationship between two variables C) Average of two variables D) Dispersion of data
- 10 If the correlation coefficient is less than the probable error, then the correlation is: A) Significant B) Reliable C) Not significant D) Perfect
- 11 Both regression coefficients cannot be: A) Positive B) Negative C) Greater than 1 simultaneously D) Equal to zero
- 12 The product of the two regression coefficients is equal to:
 A) (r) B) (r^2) C) $(1-r)$ D) $(2r)$

Q2. Solve the following.

A) Find the probability of drawing 2 red balls in succession from a bag containing 4 red and 5 black balls when the ball that is drawn first is (i) not replaced (ii) replaced.

B) In a bolt factory, three machines A, B and C manufacture 25, 35 and 40 percent of the total bolts respectively. Out of the total bolts manufactured by the machines 5, 4 and 2 percent are defective from machine A, B and C respectively. A bolt drawn at random is found to be defective. Find the probability that it was manufactured by (i) Machine A (ii) Machine A or B (iii) Machine B or C

Q3. Solve the following.

The random variable X has the following probability function

x	0	1	2	3	4	5	6	7
p(x)	0	k	2k	2k	3k	k ²	2k ²	7k ²

(i) Find the value of k. (ii) Find $P(X < 6)$, $P(X \geq 6)$ (iii) $P(0 < X < 5)$.

B) Two balls are selected at random from a box containing three red, two green and four white balls. If X and Y are the number of red balls and green balls respectively included among the two balls drawn from the box, find i) Joint probability of X and Y. ii) Marginal probabilities of X and Y. iii) Conditional distribution of X given Y=1

Q4. Solve Any Two of the following.

A die is tossed 3 times. What is the probability of (a) No fives turning up? (b) five? (c) 3 fives?

If X is binomially distributed with $E[X]=2$ and $\text{Var}[X]=4/3$, find the probability distribution of X.

A manufacture of pins knows that on an average 5% of his product is defective. He sells pins in a packet of 100 and guarantees that not more than 4 pins will be defective. What is the probability that a packet will meet the guaranteed quality? [$e^{-5} = 0.0067$]

Q5. Solve Any Two of the following.

A) Calculate the coefficient of correlation by Karl Pearson's method for the following data

x	12	9	8	10	11	13	7
y	14	9	5	2	11	12	3

B) Calculate coefficient of correlation by means of ranking method from the following data

X	40	50	60	60	80	50	70	60
Y	80	120	160	170	130	200	210	130

C) (i) Compute the correlation coefficient between the corresponding values of X and Y in the following table:

X	9	28	45	60	70	50
Y	100	60	50	40	33	57

(ii) Find the probable error.

(iii) Test the significance of the correlation.

16. Solve Any Two of the following.

- A) Calculate regression equation of Y on X and X on Y from the following data by least square method. Also calculate the value of Y when X=5. 6

X	1	2	3	4	5
Y	2	5	3	8	7

- B) Find the two regression equations from the following data. Also estimate the age of husband when the age of wife is 26. 6

Husband Age	26	28	30	31	35
Wife Age	20	27	28	30	25

Given the following data, calculate the expected value of Y when X=12. 6

	X	Y
Average	7.6	14.8
Standard deviation	3.6	2.5
r = 0.99		

End